

Optical instruments

- **Spectrometer**
- **Simple and compound microscope**
- Telescope
- Resolving power of optical instruments
- Resolving power of the eye
- Magnifying power of simple and compound microscope, telescope

Introduction

- **Optics** is an important branch of Physics, which is the study of light, its propagation, properties and other phenomena.
- Further divided into
 - **Geometrical and**
 - **Physical Optics and**
- **optical instruments** deal with the instruments which work or operated on laws and principles of one of these branches.
- Optometry is the branch of medical science that involves examining the eyes and applicable visual systems for defects

Optical Instruments

- The optical instruments are the devices which help us in seeing objects.
 - The most common examples of optical instruments are
 - human eyes,
 - microscope, and
 - telescope.
- These instruments contain lenses and form images by refraction of light through lenses.



Important Definitions Related With Eye

- **Range of Vision of a Normal Human Eye**

The range of vision of a normal human eye is from infinity of about 25 cm.

- **Near Point of human eye**

The normal human eye can focus a **sharp image** of an object on the eye if the object located any where from infinity to a certain point called the "Near Point"

- The near point of human eye is **15 cm**.

Least Distance of Distinct Vision

- The least distance of distinct vision is the minimum distance from the eyes up to which eyes can have a **clear vision** (can be seen distinctly without strain) is called least **distinct vision** of human eye
- It is about 25cm for a normal healthy **eye**.
- Thus adult eyes can see the object from **infinity to 25cm**
- **Hypermetropia** is a condition in which the eye can't see close objects properly.

Far Point of human eye

- Far point of human eye is the maximum distance or the farthest point from an eye up to which the eye can see objects clearly
- The far point of normal human eye is infinity.
- The far point of visual perception is the farthest point from which an object's image can be projected to the retina within the eye's accommodation.
- It's often referred to as the image's clearest point away from the eye.
- The far point is the limit to the eye's accommodation range The near point is the other limit of eye accommodation.
- **Myopia** is a disorder in which the eye is unable to see distant objects properly.

• **Persistence of Vision**

- When the image of an object is formed at the retina of eye then the eye continues to see the image for sometime even after the removal of object.
- This ability of human eye is called persistence of vision. In fact, the image of an object persists for $\frac{1}{16}$ th part of a second on the retina of the eye.
- The principle of persistence of vision is used in cinematography where the still pictures taken by a movie camera are projected on the screen at the rate of 24 pictures per second.

SPECTROMETER

- The spectrometer is an optical instrument used to study the **spectra of different sources of light** and to **measure the refractive indices of materials**
- It consists of basically three parts.
 - They are collimator,
 - prism table and
 - Telescope.



- The **collimator** is an arrangement to produce a parallel beam of light.
- It consists of a long cylindrical tube with a convex lens at the inner end and a vertical slit at the outer end of the tube.
- The distance between the slit and the lens can be adjusted such that the slit is at the focus of the lens.
- The slit is kept facing the source of light.
- The width of the slit can be adjusted.
- The collimator is rigidly fixed to the base of the instrument



- The **prism table** is used for mounting the prism, grating etc.
- It consists of two circular metal discs provided with three levelling screws.
- It can be rotated about a vertical axis passing through its centre and its position can be read with verniers V1 and V2.
- The prism table can be raised or lowered and can be fixed at any desired height.



- The **telescope** is an astronomical type.
- It consists of an eyepiece provided with cross wires at one end of the tube and an objective lens at its other end co-axially.
- The distance between the objective lens and the eyepiece can be adjusted so that the telescope forms a clear image at the cross wires, when a parallel beam from the collimator is incident on it.



- The telescope is attached to an arm which is capable of rotation about the same vertical axis as the prism table.
- A circular scale graduated in half degree is attached to it.
- Both the telescope and prism table are provided with radial screws for fixing them in a desired position and tangential screws for fine adjustments.



Adjustments of the spectrometer

- The following adjustments must be made before doing the experiment with spectrometer.

(i) Adjustment of the eyepiece

- The telescope is turned towards an illuminated surface and the **eyepiece is moved to and fro until the cross wires are clearly seen.**

....Adjustments of the spectrometer

(ii) Adjustment of the telescope

- The telescope is adjusted to **receive parallel rays** by turning it towards a **distant object** and adjusting the **distance between the objective lens and the eyepiece** to get a **clear image** on the cross wire.

....Adjustments of the spectrometer

(iii) Adjustment of the collimator

- **The telescope** is brought along the **axial line with the collimator**.
- The slit of the collimator is **illuminated** by a source of light.
- The distance between the **slit and the lens of the collimator is adjusted** until a clear image of the slit is seen at the cross wires of the telescope.
- Since the telescope is already adjusted for parallel rays, a **well defined image of the slit can be formed**, only when the light rays emerging from the collimator are parallel.

...Adjustments of the spectrometer

(iv) Levelling the prism table

- The prism table is **adjusted or levelled** to be in **horizontal position** by means of **levelling screws** and a **spirit level**.

Basic Function of Spectrometers

- The basic function of any spectrometer is to **take in light, break it into its spectral components, digitize the signal as a function of wavelength, and read it out and display it via a computer.**
- In **the first step** of this process, light is directed through a fiber optic cable into the spectrometer through an entrance slit, which is a narrow aperture.
- The slit pictures the light as it enters the spectrometer.
- Then, in most spectrometers, the divergent light is collimated by a concave mirror and directed onto a grating.
- Following this, the grating disperses the spectral components of the light at slightly varying angles.
- The light is then focused by a second concave mirror and imaged onto the detector.

Uses of Spectrometer

Some of the major applications of spectrometers include the following:

- used to separate, quantify, and analyze the **spectral characteristics of a light source**
- They use a grating to split light into **wavelengths of different bandwidths** and funnel that light through the spectrometer using integrated concave mirrors
- Monitoring dissolved oxygen content in freshwater and marine ecosystems
- Studying spectral emission lines of distant galaxies
- Characterization of proteins
- Space exploration
- Respiratory gas analysis in hospitals

Microscopes

- A microscope is a scientific instrument
- The term microscope can be split into two separate words, 'micro' and 'scope', where the term '**micro**' means **small or tiny**, and '**scope**' means to **view or to observe**.
- Therefore, a microscope can be understood as an instrument to observe tiny elements.
- The **optical microscope** often referred to as **the light microscope**, is a type of microscope that uses **visible light** and a system of lenses to **magnify images of small subjects**.

- Microscopes are instruments that are used in **science laboratories**, to visualize very **minute objects** such as **cells, microorganisms**, giving a **contrasting image**, that is **magnified**.
- Microscopes are made up of **lenses for magnification**, each with **their own magnification powers**.
- Depending on the **type of lens**, it will **magnify the specimen according to its focal strength**.

Types of Microscopes

- There are two basic types of optical microscopes:
 - **simple microscope** has only 1 lens.
 - **compound microscope** has 2 sets of lenses.
- It can magnify things 100 - 200 times larger than they really are.
- **Electron microscope** can magnify objects up to 300,000 times.
 - They do not use lenses, but use **electrons** to enlarge the image.

Simple Microscope



- A simple microscope is one that uses a **single lens for magnification**, such as a **magnifying glass** while a compound microscope uses several lenses to enhance the magnification of an object.
- It uses a lens to enlarge an object through angular magnification alone, giving the viewer an **erect enlarged virtual image**.
- The use of a single convex lens is found in simple magnification devices such as the **magnifying glass, loupes (used by Jewelers), and eyepieces for telescopes and microscopes**.
- It is actually a **convex lens of small focal length**, which is used for seeing the magnified images of small objects.



Principle of Simple Microscope

- A simple microscope works on the principle that when a **tiny object is placed within its focus, a virtual, erect and magnified image of the object is formed at the least distance of distinct vision from the eye held close to the lens.**

Instrumentation of Simple Microscope

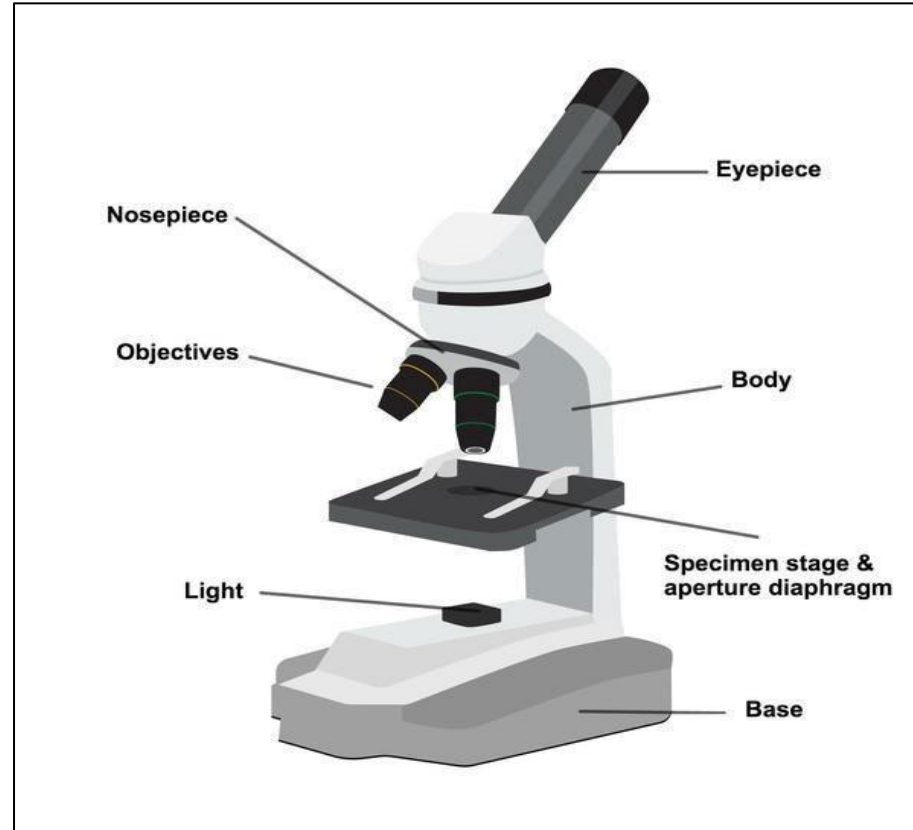


Figure: Simple Microscope Diagram

Parts of a simple microscope

- The parts of a simple microscope maybe:
 - (i) Mechanical parts
 - (ii) Optical parts

Mechanical parts:

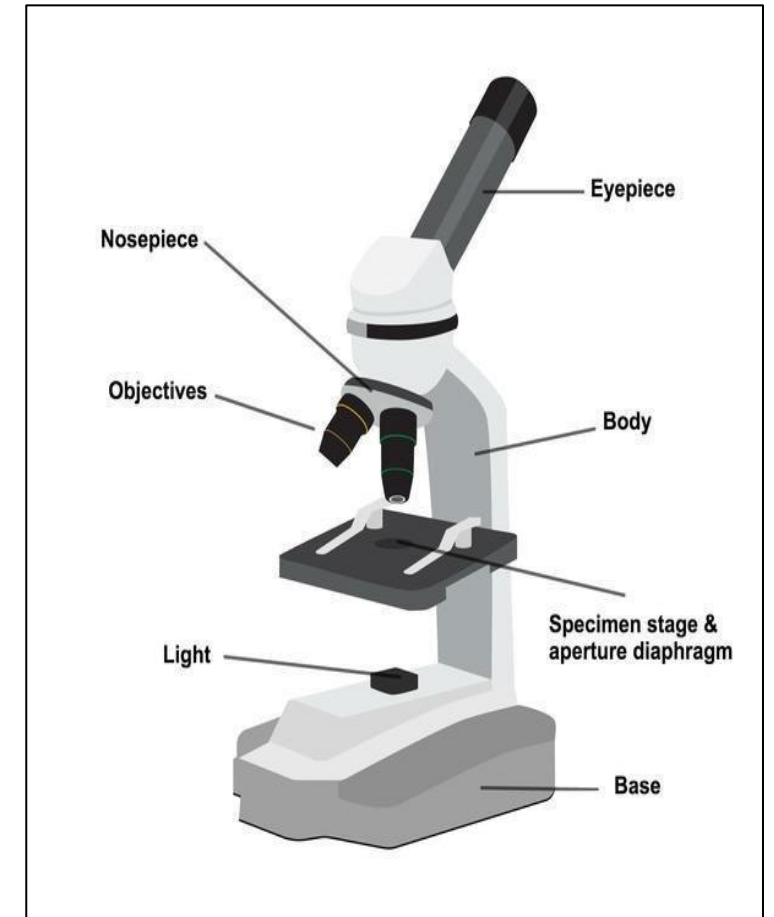
- These parts **support the optical parts** and help in their adjustment for focusing the object.
- They include the following components:

1. Metal Stand:

- It has a heavy **base plate** and a vertical rod fitted to it, which provide support and stability to other parts of the microscope.

2. Stage:

- It is a **rectangular metal plate** fitted to the vertical rod.
- It has a **central hole** for light to pass from below.
- Slide with the **specimen to be observed is kept on the stage**, in such a way that, the specimen remains just on the central hole.
- Some microscopes have a pair of slanting wings projecting from both sides of the stage. They provide support to hand for manipulating the object.



Optical parts:

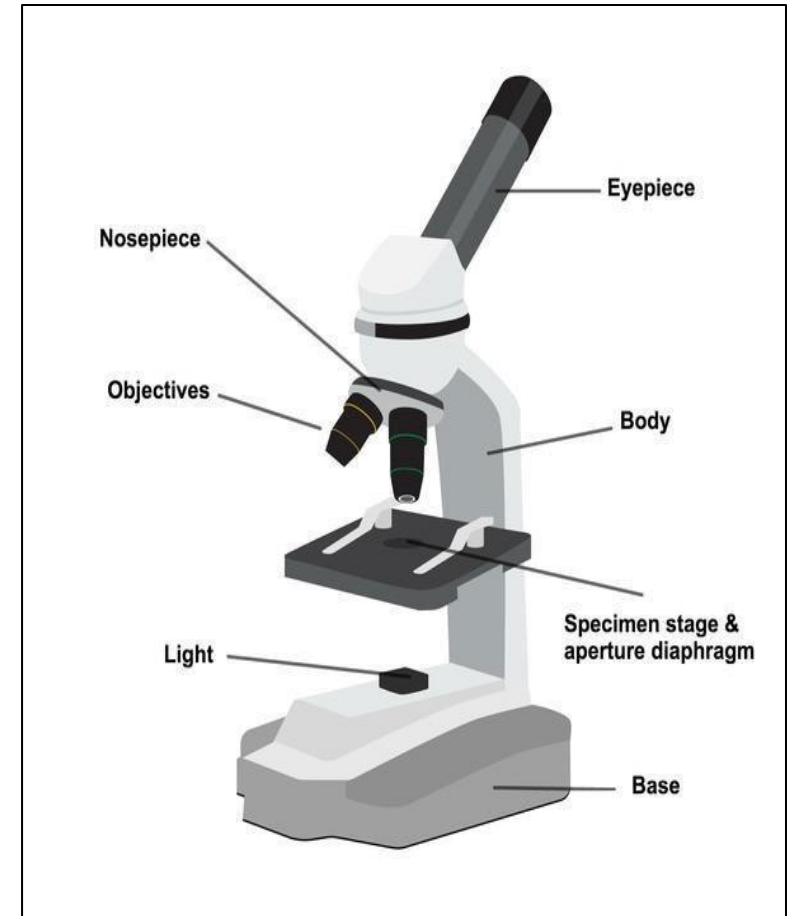
- These parts are involved in passing the light through the object (specimen) and magnifying its size.
- The components of the optical parts are as follows:

1. Mirror:

- A plano-convex mirror is fitted below the stage to the vertical rod by means of a frame.
- It focuses the **surrounding light on the object to be observed.**

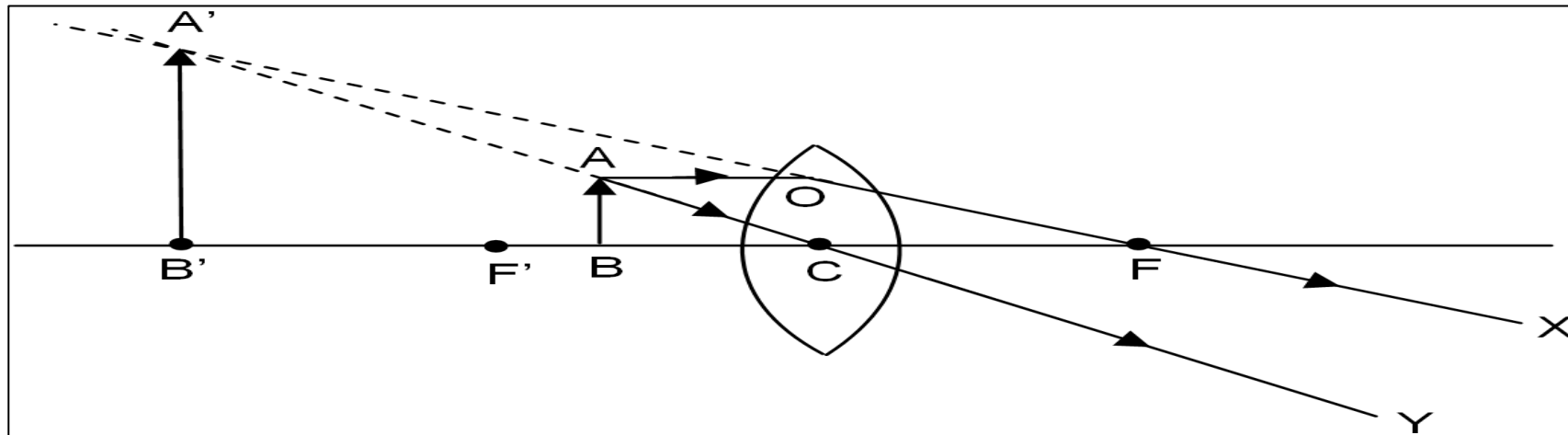
2. Lens:

- A **biconvex lens is fitted above the stage**, to the vertical rod, by means of a frame.
- It magnifies the size of the object and the enlarged virtual image formed is observed by keeping the eye above it.
- For proper focusing, the lens can be moved up and down by the frame.



Working of Simple Microscope

- The ray diagram to show the working of simple microscope is shown in figure.
- A small object AB which is to be magnified is placed **between the principal focus F' and optical centre C of the convex lens.**
- Now, a ray of light AO parallel to principal axis which is coming from the point A of the object passes through the focus F along the straight line OX after getting refracted by the convex lens.
- A second ray of light AC coming from the point A of the object passes through the optical centre C of the convex lens along the straight line CY.
- As is clear from the figure that the two rays i.e. OX and CY are diverging rays so these rays can intersect each other only at point A' when produced backward.
- Now, on drawing $A'B'$ perpendicular from point A' to the principal axis, we get the image $A'B'$ of the object which is virtual, erect and magnified.



Magnification & Resolving power

- **Magnification** : is the ability to view an object as larger.
- A good image is obtained when the amount of specimen detail is also increased.
- Magnification alone will not achieve this.
- **Resolving power** : Resolving Power is the ability to see/measure the separation of two images that are close together

Resolution	Magnification
It is possible to view 2 closely spaced objects clearly.	It is to magnify the size of objects.
It is the distinctness of 2 different objects.	It is to enlarge two different objects.
• Whenever magnification increases the resolution decreases. • Resolution à Clarity or distinctness of two different objects. • Magnification à Enlargement of the size of an object.	

Magnification of Simple Microscope

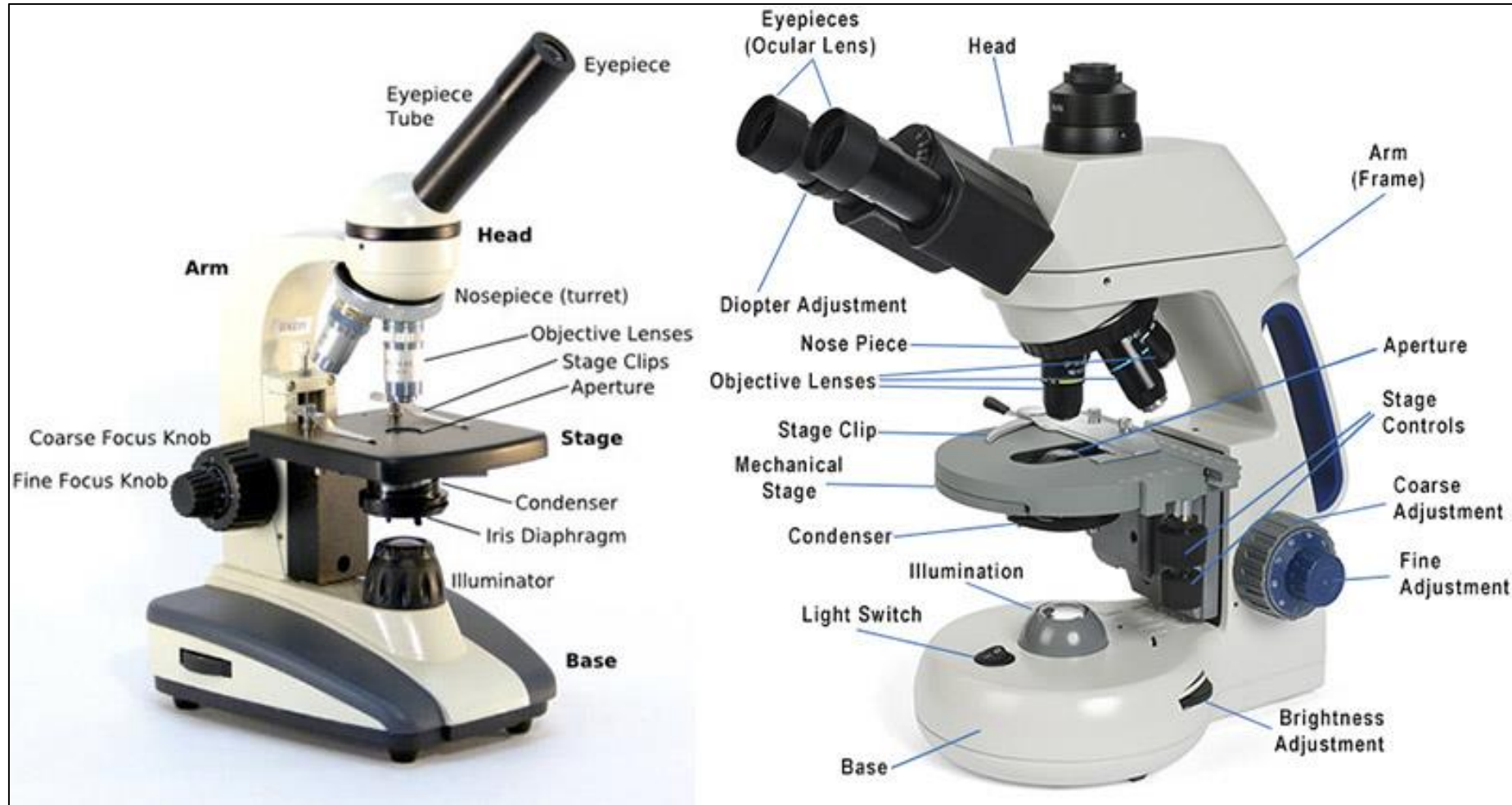
- The magnifying power of a simple microscope is given by:
- **$M = 1 + D/F$**
- Where D = the least distance of distinct vision
F = focal length of the convex lens
- The focal length of the convex lens should be small because smaller the focal length of the lens, greater will be its magnifying power.
- The maximum **magnification of a simple microscope is about 10**, which means that the object will appear **10 times larger** by using the simple microscope of maximum magnification.

Applications

- It is usually used for the study of **fungi** and **biological specimens**.
- It is commonly used by **watchmakers** to see the magnified view of small parts of a watch.
- It is also used by the jewelers to see the magnified view of the fine parts of jewelry.
- It is used to see the **enlarged image of letters of a book, textures of fibers or threads of a cloth**.
- It is used to see the magnified view of different particles of different types of **soils**.
- It is used by **palmists** to see an enlarged view of the lines of our hand.
- It is used by **skin specialists to find out various diseases of the skin**.
- It is also used to see the details of stamp and engravings.

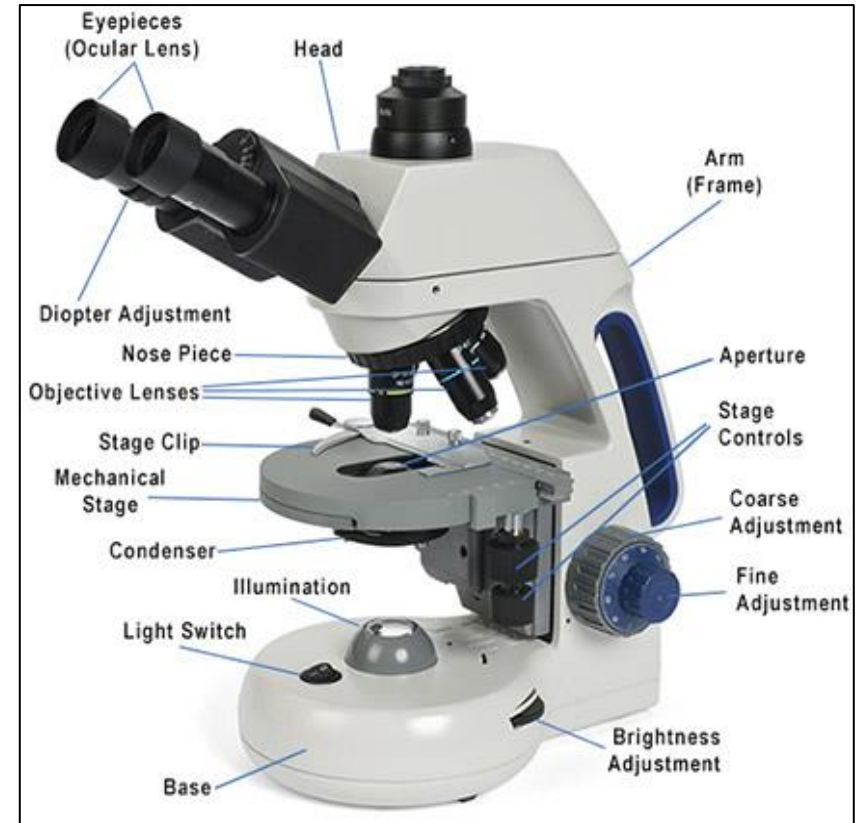
Compound microscopes

- The term “compound” in compound microscopes refers to the microscope having **more than one lens**.
- Devised with a system of **combination of lenses**, a compound microscope consists of **two optical parts**,
 - namely the **objective lens** and
 - the **ocular lens**.



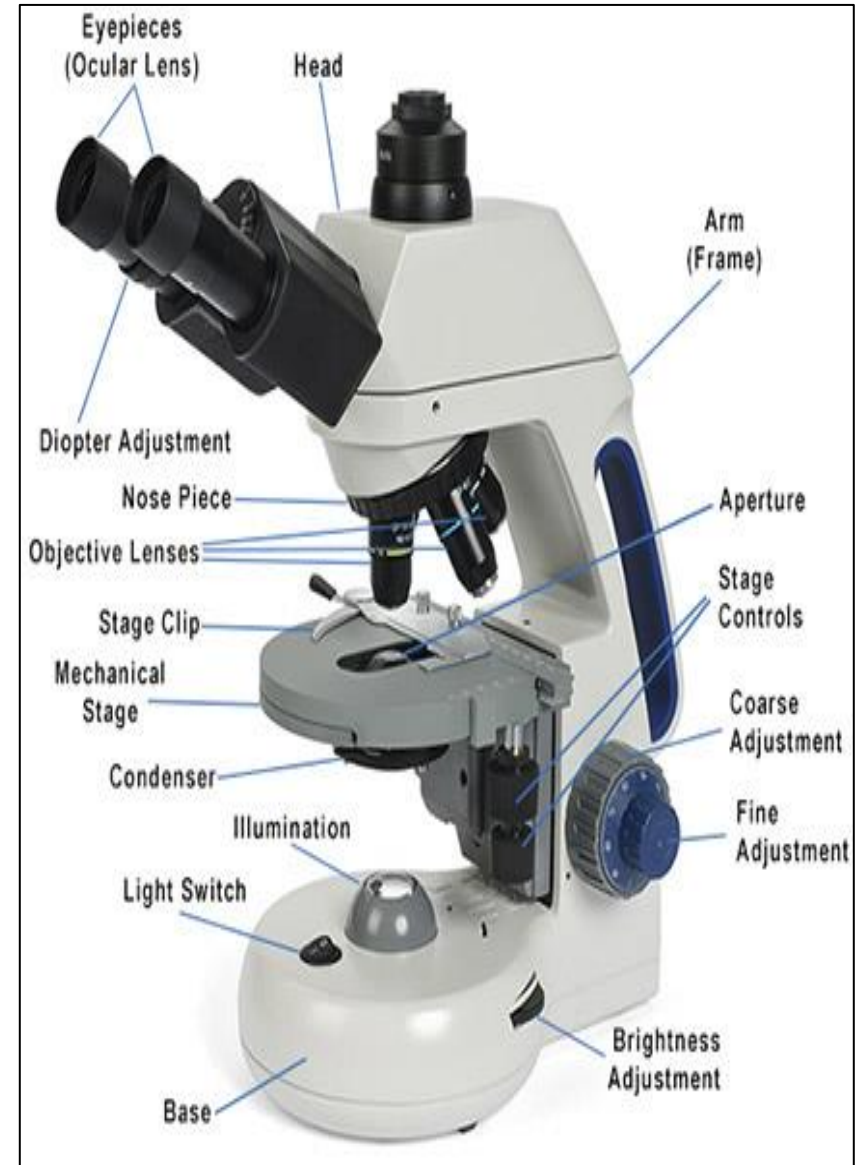
Parts of a Compound Microscope

- **Eyepiece And Body Tube.**
- The eyepiece is the lens through which the viewer looks to see the specimen.
- It usually contains a 10X (10 times closer than it does when viewed from 10" away) or 15X power lens.
- The body tube connects the eyepiece to the objective lenses.



- **Objectives and Stage Clips**

- Objective Lenses are one of the most important parts of a Compound Microscope.
- They are the closest to the specimen.
- A standard Microscope has three to four Objective Lenses which range from 4X to 100X.
- Stage Clips are metal clips that held the slide in a place.



- **Arm and Base**

- The Arm connects the Body Tube to the base of the Microscope.

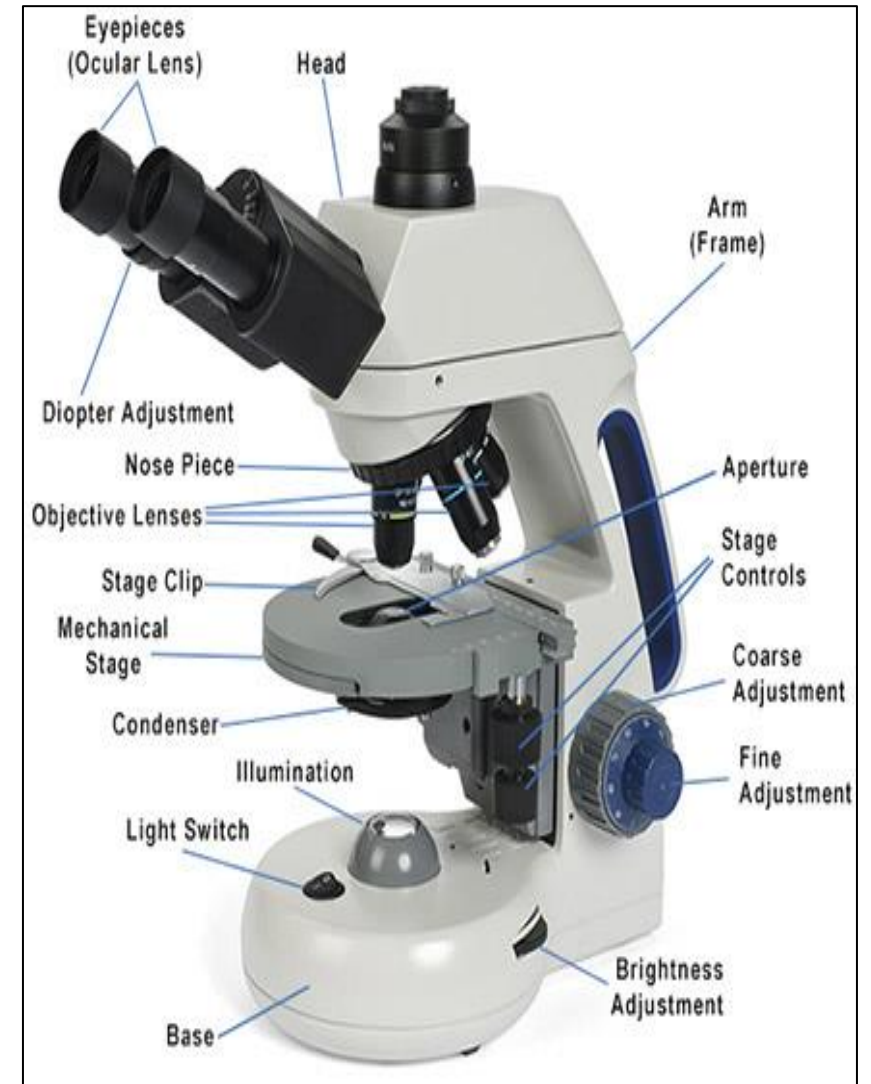
- The Base supports the Microscope and its where Illuminator.

- **Illuminator and Stage**

- The illuminator is the light source for a microscope.

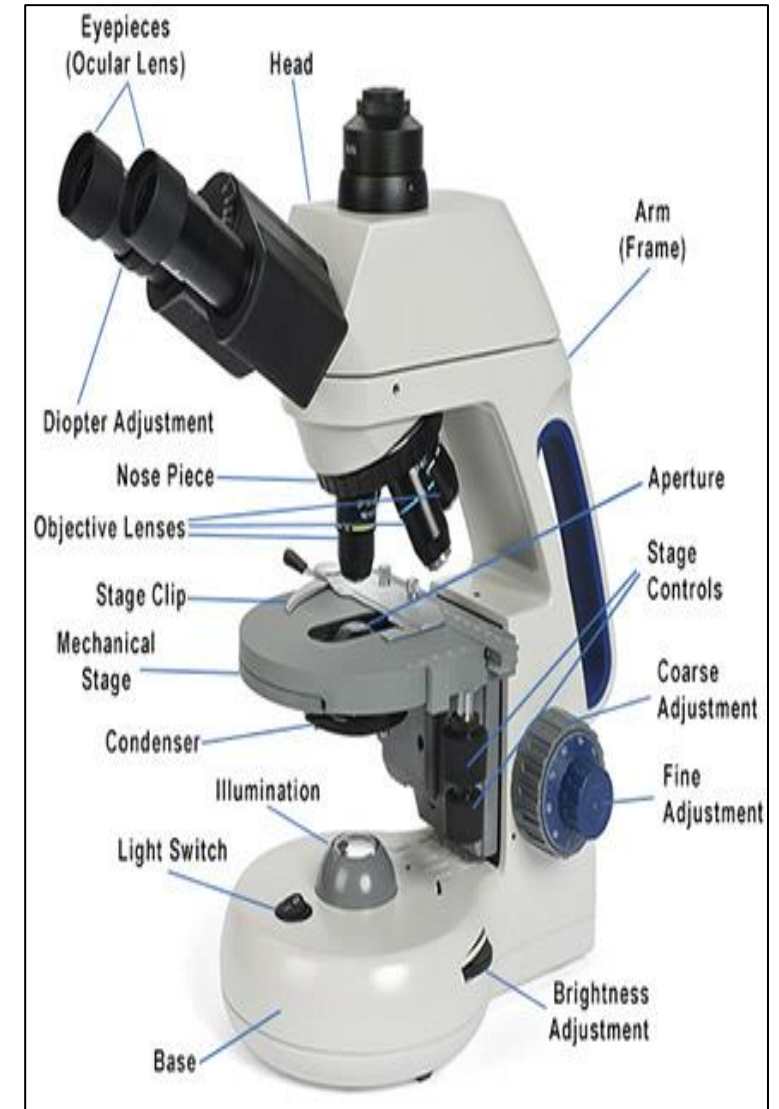
- A compound light microscope mostly uses a low voltage bulb as an illuminator.

- The stage is the flat platform where the slide is placed.



- **Nosepiece and Aperture**

- Nosepiece is a rotating tower that holds the objective lenses.
- The viewer spins the nosepiece to select different objective lenses.
- The aperture is the middle of the stage that allows light from the illuminator to reach the specimen.
- **Condenser, Iris diaphragm, and Diaphragm**
- A condenser gathers and focuses light from the illuminator onto the specimen being viewed.
- Iris diaphragm adjusts the amount of light that reaches the specimen.
- The diaphragm is a five holed disk placed under the stage
- .
- Each hole is of a different diameter. By turning it, you can vary the amount of light passing through the stage opening.

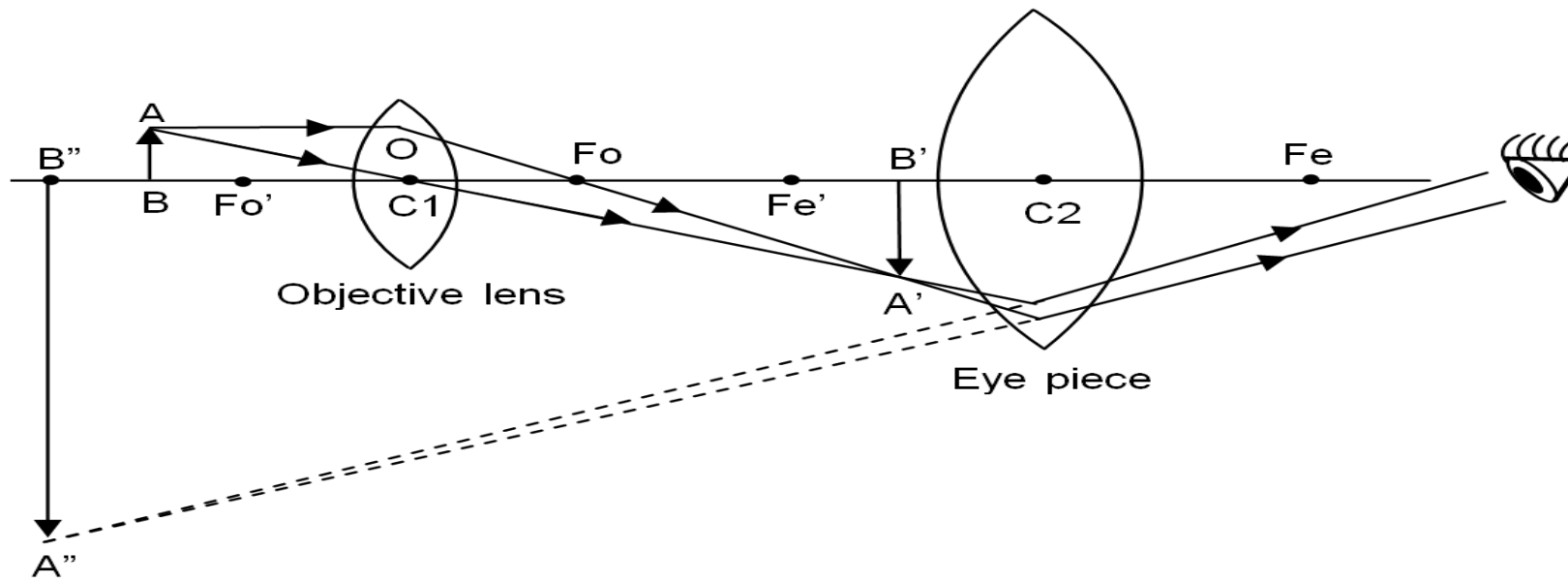


Construction of compound microscope

- A compound microscope consists of two convex lenses: an objective lens O of small aperture and an eye piece E of large aperture.
- The lens which is placed towards the object is called **objective lens**, while the lens which is towards our eye is called **eye piece**.
- These two convex lenses i.e. the objective and the eye piece **have short focal length** and are fitted at the free ends of two sliding tubes at a suitable distance from each other.
- Although the focal length of both the objective lens and eye piece is short, but the focal length of the objective lens O is a little shorter than that of the eye piece E.
- The reason for using the **eye piece of large focal length** and large aperture in a compound microscope is, so that it may **receive more light rays from the object to be magnified and form a bright image**.

Working of compound microscope

- The ray diagram to show the working of compound microscope is shown in figure.
- A tiny object AB to be magnified is placed in front of the objective lens just beyond its principal focus f_o' . In this case, the objective lens O of the compound microscope forms a real, inverted and enlarged image $A'B'$ of the object.
- Now $A'B'$ acts as an object for the eye piece E , whose position is adjusted so that $A'B'$ lies between optical centre C_2 and the focus f_e' of eye piece.
- Now the eye piece forms a final virtual, inverted and highly magnified image $A''B''$.
- this final image $A''B''$ is seen by our eye hold close to eye piece, after adjusting the final image $A''B''$ at the least distance of distinct vision of 25 cm from the eye.



Working Principle of the Compound Microscope

- A compound microscope works on the principle that when a tiny object to be magnified is placed just **beyond the focus of its objective lens**, a **virtual, inverted and highly magnified image** of the object is formed at the least distance of distinct vision from the eye held close to the eye piece
- microscopes have a combination of lenses that enhances both magnifying powers as well as the resolving power.
- The specimen or object, to be examined is usually mounted on a transparent glass slide and positioned on the specimen stage between the condenser lens and objective lens.
- A beam of visible light from the base is focused by a condenser lens onto the specimen.
- The objective lens picks up the light transmitted by the specimen and creates a magnified image of the specimen called the primary image inside the body tube. This image is again magnified by the ocular lens or eyepiece.

Working Mechanism Of The Compound Microscope

- View into the eyepiece. Rearrange the mirror such that adequate light passes into the microscope
- The mirror, lenses, stage, and slides should be cleared of dust and be clean.
- Place the slide in the middle of the stage
- Firmly secure the slide with clips at two edges of the slide to ensure that the slide cannot move
- The nose piece is adjusted in such a way that the low power objective is aligned with the object of focus placed on the slide.
- The coarse adjustment knob can be shifted upwards or downwards such that the slide is well under focus
- Turn the fine adjustment knob by moving upwards or downwards to get a clear and sharp image of the object under focus.
- All minute details of the object are observed under low power objective.
- The nose piece is now turned to bring the high power objective aligning with the object.
- The fine adjustment knob is tuned as much as possible to get a bright and precise view of the object.
- In high power, the details of the object are observed.
- The coarse adjustment knob should not be used when the object is being examined in high power as it can crush the slide.

Magnification of compound microscope

- In order to ascertain the total magnification when viewing an image with a compound light microscope, take the power of the objective lens which is at 4x, 10x or 40x and multiply it by the power of the eyepiece which is typically **10x**.
- Therefore, a 10x eyepiece used with a 40X objective lens will produce a magnification of 400X.
- The naked eye can now view the specimen at magnification 400 times greater and so microscopic details are revealed.

- Alternatively, the magnification of the compound microscope is given by:

- $$m = D / f_o * L / f_e$$

- where, D = Least distance of distinct vision (25 cm)
L = Length of the microscope tube
fo = Focal length of the objective lens
fe = Focal length of the eye-piece lens

Applications

- A compound microscope is of great use in pathology labs so as to identify diseases.
- Various crime cases are detected and solved by drawing out human cells and examining them under the microscope in forensic laboratories.
- The presence or absence of minerals and the presence of metals can be identified using compound microscopes.
- Students in schools and colleges are benefited by the use of a microscope for conducting their academic experiments.
- It helps to see and understand the microbial world of bacteria and viruses, which is otherwise invisible to the naked eye.
- Plant cells are examined and the microorganisms thriving on it can be ascertained with the help of a compound microscope. Thereby, a compound microscope has proved to be crucial to biologists.

Advantages

- Simplicity and its convenience.
- A compound light microscope is relatively small, therefore it's easy to use and simple to store, and it comes with its own light source.
- Because of their multiple lenses, compound light microscopes are able to reveal a great amount of detail in samples.

Precautions

- The objective lenses and eyepiece should be cleaned with the help of silk cloth and cleaning liquid before using.
- The microscope should not be tilted when working, using it.
- When an object needs to be studied, focus on low power objective first and then move to high power.
- The lower power needs to be left in place after all observations are completed.
- When focusing care needs to be taken to ensure that the objective lens never strikes the stage or the slide.
- Only the fine adjustment knob should be used when the high power objective is employed.

...Precautions

- Cover slip should always be used to cover well-mount preparations before observation under the microscope is made.
- Do not dismantle the microscope.
- When carrying the microscope, always use both hands.
- Try to place it in a box after using the microscope.
- The concave part of the mirror should be used under dim light.
- Oil immersion lens should never be used without the use of oil.

Differences between Simple Microscope and Compound Microscope

Simple Microscope	Compound Microscope
A simple microscope is also called magnifying glass. It is actually a convex lens of small focal length, which is used for seeing the magnified images of small objects.	A compound microscope is an optical instrument consisting of two convex lenses of short focal lengths which is used for observing the highly magnified images of tiny objects.
It consists of one convex lens.	It consists of two convex lenses of short focal length.
Its maximum magnifying power is 10.	Its maximum magnifying power is 1000.

Characteristics	Simple Microscope	Compound Microscope
Number of lenses	One to magnify objects	3-5 to magnify objects
Condenser lens	Absent	Present
Light source	Natural	Illuminator
Mirror type	Concave reflecting	One side is plain and the other side is concave
Level of magnification	One level only	Higher level
Magnifying power	Up to 300X	2,000X
Adjusting Magnification	No	Yes
Used	At basic level	By professional for research purpose

Thank you